

Highlights

Cooling Materials for Sustainable Compute Waste-Heat Recovery (Bustamante, RSER 2026 submission)

- Materials-outward review of three generations of compute cooling for waste-heat reuse
- Carnot exergy of recovered heat doubles, then triples, across the three generations
- Breakeven matrix: mining-as-heating works only on grids below $\sim 0.13\text{--}0.24$ tCO₂/MWh
- Parallel cold-plate AI case: net-positive in Finland, net-negative in Germany
- 17 operational heat-reuse deployments documented across 9 countries, 3 continents

Each highlight is a maximum of 85 characters including spaces, per RSER author guidelines.